PSYC4411

FIND STRUCTURE, DON'T FABRICATE IT

WEEK 8 CHALLENGE LAB

PCA, UMAP & Clustering on Real Mental Health Data

Monday 27 April is a public holiday — complete this lab in your assigned groups in your own time

TODAY'S CHALLENGE

- **Phase 1: Dimensionality Reduction**
  - PCA on 42 DASS items — how many components?
  - UMAP visualisation coloured by depression severity
- **Phase 2: Clustering**
  - k-Means on Big Five + DASS subscales (8 features)
  - Profile your clusters — what do they mean psychologically?
- **Stability & Documentation**
  - Test whether your clusters survive a different random seed
  - Write a methods paragraph — then verify the AI got it right
- **Prepare a summary slide**
  - Scree plot, UMAP, cluster profiles, stability check, methods paragraph correction

THE DATASET: DASS-42 (FROM WEEK 4)

- **Same data, different question**
  - Week 4: "Can personality predict depression?"
  - Week 8: "Are there subgroups in how people experience distress?"
- **After cleaning:** ~34,500 respondents
- **PCA input:** 42 DASS items (distress symptoms)
- **Clustering input:** 5 Big Five + 3 DASS subscales = 8 features

Data prep is done for you! The starter notebook loads, cleans, and scores the data in cells 1–5. You start the analysis from cell 6.

New this week: VCL fake word filter removes careless responders — participants who claim to know words that don't exist.

NEW LLM SKILL: DOCUMENTATION

Week 2: **Prompting** · Week 4: **Debugging** · Week 6: **Refactoring** · Week 8: **Documentation**

Weak

"Explain my code."

Strong

"I ran PCA on 42 DASS items from 34,500 participants. Write a methods paragraph for a psychology journal (APA style). Include: sample size, measures, preprocessing, number of components, variance explained, clustering details, evaluation metrics. Be specific about software."

The AI's draft is a starting point. You MUST verify every number and method name matches your actual analysis. Documentation that doesn't match is worse than no documentation.

GETTING STARTED

STEPS

1. Meet with your **assigned group** (in person or online)
2. Open `starter.ipynb` or `starter.py`
3. Run cells 1–5 (data is loaded, cleaned & scored)
4. Ask your AI to **plan first**
5. Phase 1: PCA → loadings → UMAP
6. Phase 2: Cluster → profile → stability
7. Write methods paragraph & prepare summary slide

TARGET NUMBERS

PCA: 3 clear components, ~55% variance

PC1: ~45% — general distress factor

Best silhouette: k=2 (~0.24)

Stability: very high for k=2 (<1% change)